

VISHAL GUPTA



DevOps | SRE

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PROFESSIONAL SUMMARY

AWS Certified DevOps Engineer | SRE with 6+ Years of exp. in managing infrastructure in multi-cloud ecosystem with main focus on Automation, Scalability, Reliability, and Performance. Proficient in designing resilient architectures, automating infrastructure and deployments, optimizing cost, and observability. Dedicated to fostering a culture of reliability and continuous improvement.

TECHNICAL EXPERTISE

SCM: Bitbucket, GitHub, Gitlab, Azure Git	OS: CentOS, RHEL, Ubuntu, Windows, Mac
Cloud Computing: AWS, GCP and Azure	Scripting: Shell, Powershell, Python, YAML
Orchestration: Kubernetes, OpenShift	Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch
Virtualization Tools: Docker, Virtual Box, Vagrant	Web Servers: Nginx and Apache
IAAC Tools: Terraform, CloudFormation, ARM Templates, Packer	Configuration Management Tool: Ansible, Salt, Chef
Build Automation Tools: Jenkins, AWS CodePipeline, Gitlab CI, Azure DevOps, ArgoCD, FluxCD	Code Quality Test: SonarQube, Checkov
Monitoring, Logging, Alerting: Datadog, Victoria Metrics, Grafana, ELK, Fluentd, Fluentbit, PagerDuty	Linux: HTTP, ISCSI, FTP, NFS, DNS, SSH, LVM, LDAP, ACL, Cronjobs, Permissions, Monit
Enterprise Tools: Morpheus, CommVault, Terraform Enterprise, RedHat Satellite, Cloudflare, Spot.io, Sprinto	Other Tools: Jira, Nexus, Active Directory, ServiceNow, Apache Hadoop, Chart Museum, Solr, HAProxy, OAuthProxy, SOC2 Framework, Argo Workflow, Kubeflow

GLOBAL CERTIFICATIONS

- AWS Certified DevOps Engineer - Professional Cert ID: 7T504P5CH1BQ1T56
- AWS Certified Solutions Architect Associate (AWS CSA) Cert ID: S969ZM1CMFEE1WK8
- AWS Technical Professional (Digital)
- Certified Kubernetes Administrator Cert ID: LF-tefyeo025k
- Red Hat Certified System Administrator (RHCSA) Cert ID: 180-133-423
- Red Hat Specialist in Ansible Automation (RHCOE in Ansible Automation) Cert ID: 180-133-423
- Red Hat Specialist in OpenShift Administration (RHCOE in OpenShift Administration) Cert ID: 180-133-42

PROFESSIONAL WORK EXPERIENCE



DEVOPS ENGINEER - III – NETCORE CLOUD

May 2021 - Present

- Migrated and Orchestrated 150+ microservices from VMs to multi-region Kubernetes Self Hosted KOPS, EKS and GKE. **Achieved a ~40% reduction** in infrastructure costs through optimized resource management and strategic provisioning.
- Upgrades:** Facilitated EKS and GKE version Upgrades with zero downtime using Node Pool Blue-Green deployment strategy.
- Designed and automated **one-click Deployment solutions for cross-functional teams** across multi-regions AWS and GCP Kubernetes clusters using Terraform and Flux modules, supporting workloads for Solr, Feed, Platform, SelfServe, PIM, and Console.
 - Implemented Config Mgmt through IAAC **Terraform modules**, incorporating components like HPA Ingress Controllers, Logging(EFK)-Monitoring(VM, Grafana) components, EFS-FSx CSI drivers, Filestores, Storage Classes, PV PVCs, EKS AddOns, ArgoCD, Argo Workflows, Solr, ImagePullSecrets-Patcher to SA, external DNS, spot.io ocean controllers, and spot.io Virtual Node Groups.
 - Spearheaded the setup of a centralized CI/CD environment across AWS and GCP using **FluxCD GitOps**, optimizing deployment workflows with GitOps, Kustomization CRDs, Helm releases, and release orchestration.
- GCP Migration:** Automated VM-based deployments for handling multi-region capability using an Immutable infrastructure model using Jenkins, Salt, Packer & Terraform.
 - Stateless App Migration: Successfully migrated applications from AWS to GCP without any downtime, ensuring continuous availability.
 - Custom Helm Charts: Developed custom Helm charts to deploy applications in production, ensuring adherence to best practices.
 - Automation: Utilized SaltStack for VM configuration management and Packer for automated image creation for GC. Provisioned GCP resources with Terraform, automating VM lifecycles, networks, and storage configurations.
 - Jenkins: Developed Jenkins CI/CD pipelines for seamless build of Stateful applications across regions.
 - GitOps with FluxCD: Employed a GitOps approach to deploy applications on GKE clusters using FluxCD.
 - DNS Migration: Managed the migration of AWS Route 53 to GCP Cloud DNS and Cloudflare, ensuring seamless domain management.
 - Database Migration: Managed migration of stateful services like Elasticsearch, MongoDB, Apache Storm to GCP with minimal downtime using a Blue-Green deployment strategy.
 - CIDR Planning: Strategically planned CIDR allocations to prevent conflicts during the creation of multiple networks and migration activities.
 - SSO: Deployed Argo Workflows, Implemented RBAC, Single Sign-On (SSO) using GitHub OAuth and Dex for secure access to the workflow management system.
- Configured **inter-cloud communication** across 10 AWS VPCs and 3 GCP regions using Site-to-Site VPN and Transit Gateways. Configured networking in K8s envs to access private endpoints hosted in AWS and GCP.
- Designed and setup robust highly available **centralized EFK logging architecture** handling 20TB of logs daily for multi-cloud environments.
- Set up **Monitoring** dashboards and alerts using Victoria Metrics, Grafana, Datadog, and PagerDuty, ensuring performance visibility for VMs and k8s clusters across all regions.
- Implemented an end-to-end high-security control framework for **SOC2 compliance**, improving system security and managing vulnerabilities.
- Cost Optimization:** Implemented strategies to optimize server costs, resulting in significant savings.
 - Spot Instances and Pre-emptible Instances: Utilized spot and pre-emptible instances to reduce costs.
 - VPC Endpoint Usage: Ensured read and write operations were over private routes using VPC endpoints to avoid NAT gateway costs.
 - Custom Downscaling Script: Developed scripts to downscale instances during low-utilization periods, ensuring priority applications are not impacted.
 - S3, ECR Lifecycle Rules: Implemented lifecycle rules to transition objects to Glacier and Glacier Deep Archive, saving costs. and established retention policies to delete unwanted data.
 - Cost Alerts: Created alerts to monitor sudden cost increases, enabling quick troubleshooting and resolution.
 - Graviton-Based Migrations: Led migration initiatives to Graviton-based instances for better performance and cost efficiency.
 - Ingress Adoption: Adopted efficient ingress strategies to reduce network costs and improve service delivery.
 - Service-to-Service Communication: Enabled service-to-service communication within the cluster using private routes via CoreDNS, significantly reducing load balancer costs.
 - GP3 Volumes: Integrated CSI driver to launch GP3 volumes for Kubernetes PVs, root volumes, and all EBS volumes across EC2 instances.
 - EC2, EBS, ElastiCache RDS Optimization: Identified and eliminated underutilized or unused EC2 instances, EIP, RDS instances, EBS volumes, aged RDS snapshots to improve cost efficiency.
- BAU and On-call: Consistently delivered business-as-usual support and maintained on-call readiness to address urgent issues.
- Readiness and Support: Prepared and supported Mega Blockbuster, Black Friday, Cyber Monday, and Diwali sale, ensuring they met operational requirements.
- Disaster Recovery (DR) Project: Full Infrastructure for PO Applications: Created a comprehensive infrastructure for priority 0 (P0) applications to ensure availability during major incidents such as region or zone outages, database failures, PO services disruptions, and CDN failures.
- Cloudflare Zero Trust:** Secured SSH access to Jump Servers through Cloudflare Proxy with 2FA authentication.

- o Created Cloudflare Access policies for user authentication (only @netcoreunbx.com emails).
 - o Configured SSH to route through Cloudflare, eliminating open SSH ports.
 - o Set up Cloudflare Tunnel with DNS routes and Systemd service for persistence.
 - o Implemented secure, scalable SSH access, eliminating direct EC2 access and enhancing security.
- Ownership and Proactive Solutions: Demonstrated dedication and expertise in overcoming critical challenges, taking ownership of problems, proactively finding solutions, acted as the go-to person for debugging and troubleshooting production latency/performance issues.
- Set up automatic maintenance and repair of Linux processes with Monit; managed legacy apps using Chef/Opsworks for workflow automation.
- Developed automation solutions via Shell/Python Scripts, Ansible to improve and support the productivity of the Unbx applications.
- Automated etcd-manager upgrades and rolling updates in KOPS clusters, ensuring TLS certificate rotation and preventing kube-api downtime. Upgraded etcd to v3.0.2+ for automated certificate management.
- Able to demonstrate proficiency in 3 cloud services and able to rapidly incorporate new technologies as required.
- **Technologies used - Kubernetes (KOPS, EKS, GKE), Blue Green K8s Upgrades, Jenkins, Salt, Packer, Terraform, Chef, OpsWorks, Solr, Fluentbit, Fluentd, Elasticsearch, Kibana, Victoria Metrics, Grafana, Datadog, PagerDuty, Monit, SOC2 Compliance Framework, AWS, GCP, Cloudflare, ArgoCD, FluxCD Argo Workflows, Kubeflow.**



SENIOR DEVOPS ENGINEER – HASHEDIN BY DELOITTE

Jan 2019 - May 2021

PROJECT: UNISYS CLOUDFORTE CLOUD MANAGEMENT PLATFORM (CMP)

- Leveraged Morpheus (Agnostic Hybrid Cloud Management Platform) to create Blueprints, Tasks, Workflows, and Apps for AWS, Azure, GCP, and VMware, automating cloud resource provisioning, configuration, and scaling with Ansible and cloud-native tools, reducing deployment time by 40%.
- Configured VPN tunnel connectivity between Azure VNETs, AWS VPCs, and GCP VPCs to enable secure inter-VPC and VNet communication between private subnets.
- Registered AWS, Azure, GCP and VMware Instances to Windows Active Directory Domain using Powershell and Shell Scripts such that instances can be logged in via users from Windows Active Directory.
- Cloud Enhancement and Extension Services: Automated instance provisioning across AWS, Azure, GKE, VMware, and Oracle Clouds for streamlined cloud operations.
 - o Automated Patching Solution: Integrated WinSus and AWS Patch Management with Morpheus Blueprint, ensuring instances are registered to AWS Systems Manager and patches are applied during maintenance windows.
 - o Automated Monitoring Solution: Deployed Vanda and AWS CloudWatch for monitoring Windows and Linux instances, using custom scripts to track system metrics and set alarms.
 - o Automated Backup Solution: Configured backup policies using CommVault and AWS Backup Service, centralizing backup monitoring and management for AWS resources.
- CMP CI/CD: Configured Jenkins build machine for CMP, integrated with SCM, automated WHL file creation from Python scripts, and uploaded build artifacts to Nexus repository.
 - o Automated Terraform-based Azure provisioning, installed Morpheus on Linux VMs, and ran Python scripts to import CMP configurations.
 - o Used Checkov to scan Terraform-managed cloud infrastructure, ensuring security and compliance best practices.
 - o Executed Python automation scripts for sanity testing on Windows nodes, passing appliance URLs via Jenkins.
 - o Set up email approval for Terraform resource destruction and automated workspace cleanup in Jenkins.
- Created Ansible playbooks for package installation and Kerberos authentication on Windows and Linux servers (Ubuntu, RHEL, CentOS, Amazon Linux), used Ansible Vault for secure data storage and integrated with Morpheus.
- Implemented Azure AD SAML-based Single Sign-On (SSO) for the Morpheus platform, integrating it with an enterprise application in Azure to enable seamless authentication, streamline user access with automated role-based access control (RBAC), and ensure compliance with GDPR and HIPAA standards.
- Developed Shell and PowerShell scripts for task cleanup, logging aggregation, error handling, and immediate patch installation on Windows and Linux instances post-provisioning.
- SSL Certificates encryption to the Morpheus Nginx Servers.
- Implemented server hardening in Unisys environment by creating Bastion hosts, restricting ports, and configuring load balancers for secure 443 traffic. Enabled GuardDuty, AWS Config, CloudTrail, Security Hub, and CloudWatch for continuous monitoring, resulting in a 30% reduction in security incidents.
- Achieved a 15-20% reduction in cloud expenditures by identifying and addressing underutilized infrastructure.
- **Technologies used - Morpheus, Ansible, AWS, Azure, GCP, Shell Script, PowerShell Script, Jenkins, Kubernetes**

PROJECT: BRANDIX MULTI-ACCOUNT AWS NETWORKING SETUP

- AWS Account Setup & VPC Design: Automated the creation of 7 AWS accounts using Account Vending Machine (AVM), AWS Organizations, and Organizational Units (OU). Established inter-account VPC, VPN, and internet connectivity between the accounts. Configured private subnets for 6 accounts and a public subnet in the Network Account.
- Transit Gateway & VPN Integration: Configured AWS Transit Gateway (TGW) for cross-account connectivity, shared via AWS Resource Access Manager (RAM). Established Site-to-Site VPN with Customer and Virtual Private Gateways (VGW), ensuring secure, redundant IPsec VPN tunnels for on-premise to AWS communication.
- EKS Cluster Deployment & Security: Deployed EKS with Node Autoscaling in Shared Service Dev Account and implemented EFK Fluentbit DaemonSet for centralized logging. Configured IAM roles, GuardDuty, AWS SSO, KMS, Security Hub, CloudTrail, and CloudWatch for security and compliance across all AWS accounts.
- Created Cloudwatch Event Rules to trigger Lambda function on a schedule to stop and start Amazon EC2 instances at regular intervals.
- **Technologies used - AWS Organizations, Transit Gateway, Amazon VPC, Amazon EKS, GuardDuty, AWS SSO, AWS KMS, Security Hub, CloudTrail, AWS Config, CloudWatch, PowerShell scripting, AWS Lambda.**

PROJECT: SWIGGY, JITFIN, DIGID8 KUBERNETES, ELASTIC BEANSTALK DEPLOYMENT

- Automated Infrastructure Provisioning: Automated Provisioning of VPC with public/private subnets, EC2, ASG, LB, SG, Route 53 Records, S3 Buckets, CloudFront, RDS PostgreSQL database Elastic Beanstalk, and EKS using Terraform for Dev, QA, and Prod environments. Containerized a three-tier application with best practices and security measures.
- CI/CD Pipeline Development: Developed a multi-stage CI/CD pipeline for Frontend releases in S3, CloudFront, and Backend deployment in Elastic Beanstalk/EKS.
- Nginx Web Server & SES Integration: Configured Nginx for reverse proxy and static file serving. Set up SES for email delivery using SMTP, integrated with SNS for pipeline execution alerts via CloudWatch Event Rules.
- Monitoring & Logging Integration: Integrated Filebeat, ELK Stack for Docker log aggregation, and used AWS CloudWatch for infrastructure monitoring with custom metrics.
- Secrets Management & Data Security: Managed sensitive environment variables with AWS SSM Parameter Store for secure storage and access control.
- **Technologies: AWS, EKS, Docker, Terraform, Kubernetes, Nginx, SNS, SES, SSM.**



DEVOPS INTERN – LINUX WORLD PVT. LTD.

Jun 2018 - Dec 2018

- Established High Availability Hadoop Cluster on AWS and Vagrant. Automated IAAC and Config Mgmt via Ansible.
- Enabled automatic failover and data protection via Apache Zookeeper.
- Streamlined workflow with a CI/CD Jenkins pipeline for EC2 provisioning and High Availability Hadoop setup on Vagrant machines and AWS cloud.
- **Technologies used: Hadoop, Zookeeper, AWS, Vagrant, Ansible, Jenkins**

EDUCATION

JK LAKSHMIPAT UNIVERSITY

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE ENGINEERING

Grade - 8.5

Awarded Scholarship for eight semesters. Delivered Sessions on Linux and Docker as Technical Head of CSI JKLU.